AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI

ANNUAL EXAMINATIONS 2021

Computer Science

Total Time: 2 hours 10 minutes

Total Marks: 65 (50-Theory & 15-Alternate to Practical)

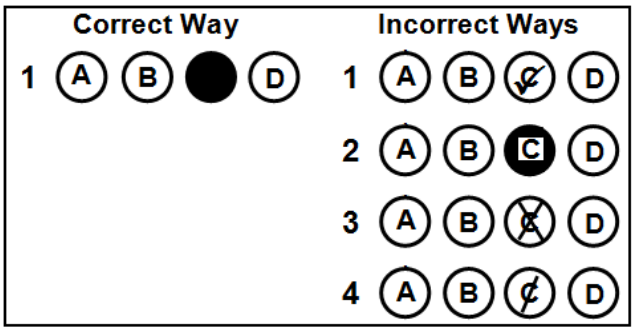
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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 65 only.
4. Question Distribution:

Theory	Alternate to Practical (ATP)
50 MCQs	15 MCQs

5. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

6. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
7. DO NOT write anything in the answer grid. The computer only records what is in the circles.
8. The marks obtained on the 50 MCQs will be equated to the total marks of 85 for the theory examination results.
9. You may use a scientific calculator if you wish.

THEORY (Questions 1-50)

1. Examples of impact printers are

   A. dot matrix and inkjet printers.
   B. dot matrix and chain printers.
   C. laser and chain printers.
   D. laser and inkjet printers.

2. Humera has handwritten text on her diary. She wants to store this text in digital form on computer. The device that can help her is

   A. mouse.
   B. printer.
   C. scanner.
   D. touchpad.

3. The MOST common built-in pointing device in a laptop is

   A. mouse.
   B. trackball.
   C. light pen.
   D. touchpad.

4. An example of open source software is

   A. Skype.
   B. Linux.
   C. Adobe Reader.
   D. Microsoft Windows.

5. Which of the following stages requires arithmetic logic unit (ALU)?

   A. Input
   B. Output
   C. Storage
   D. Process

6. Which of the following is the CORRECT categorisation of manual and automatic data capture devices?

	Manual Data Capture Devices	**Automatic Data Capture Devices**
A	Scanner	Mouse
B	Keyboard	Magnetic Card Reader
C	Barcode Reader	Graphics Tablet
D	Quick Response Code Reader	Optical Character Reader

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7. An example of a computing device is

   A. router.
   B. mouse.
   C. touchpad.
   D. microphone.

8. Which of the following computers are MOSTLY used in industrial process control?

   A. Minicomputers
   B. Microcomputers
   C. Supercomputers
   D. Mainframe Computers

9. A software that reduces the logical size of a file to save disk space is an example of

   A. device driver.
   B. utility software.
   C. operating system.
   D. productivity software.

10. Which of the following devices sends analog signals to computer that need to be converted into digital before processor can process them?

    A. Speaker
    B. Touchpad
    C. Keyboard
    D. Microphone

11. Read the given statements related to inkjet and laser printers.

    I. These printers use paper feed.
    II. These printers use dry powder ink.
    III. These printers print the whole page in one go.
    IV. The print head of these printers consists of nozzles.

    Which of the following options shows the CORRECT categorisation of these statements?

	Inkjet Printer	**Laser Printer**
A	I and II	III and IV
B	I and IV	II and III
C	III and IV	I and II
D	II and III	I and IV

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12. The instruction given by a user to defrost food in a microwave oven for a specific time is stored in

A. RAM.
B. ROM.
C. hard drive.
D. flash drive.

13. Which of the following options shows the CORRECT categorisation of computers based on their speed?

	Low Speed ⟶		**High Speed**
A	Mainframe Computer	Minicomputer	Microcomputer
B	Microcomputer	Minicomputer	Mainframe Computer
C	Minicomputer	Microcomputer	Mainframe Computer
D	Microcomputer	Mainframe Computer	Minicomputer

14. The MOST suitable computer system to show a video via projector is the

A. minicomputer.
B. supercomputer.
C. microcomputer.
D. mainframe computer.

15. The number of kibibytes in 0.50 gibibyte is

A. 128
B. 512
C. 131,072
D. 524,288

16. A CPU directly reads from and writes to a memory. The name of this memory is

A. RAM.
B. ROM.
C. hard disk.
D. compact disk.

17. The MAIN characteristic due to which SRAM is used as cache memory in computers is that it

A. is fast.
B. is cheap.
C. is refreshed periodically.
D. uses less number of transistors.

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18. Examples of primary memory are

A. cache and ROM.
B. SD card and DVD.
C. RAM and hard disk.
D. cache and flash drive.

19. Omar stored 5 GB data on an optical storage media. Later on, he stored 3 GB more data on the same optical storage media.

The type of optical storage he used is

A. DVD-R
B. DVD-RW
C. CD-ROM
D. DVD-ROM

20. Which of the following memories, a CPU can directly communicate with?

A. Register
B. CD-ROM
C. DVD-ROM
D. USB Flash Drive

21. A program that performs power on self-test (POST) is stored in the

A. register.
B. hard disk.
C. cache memory.
D. read only memory.

22. A memory device having the smallest storage capacity is the

A. register.
B. compact disk.
C. cache memory.
D. random access memory.

23. Which of the following statements is TRUE about DRAM?

A. It consists of vacuum tubes and processors.
B. It consists of vacuum tubes and transistors.
C. It consists of capacitors and processors.
D. It consists of transistors and capacitors.

24. How many gigabytes are there in 2 petabytes?

A. 2,400,000
B. 2,000,000
C. 2,400
D. 2,000

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25. In contrast to the primary memory, the secondary memory

A. is costly.
B. is volatile in nature.
C. is accessed by I/O channels.
D. always has lower storage capacity.

26. Which of the following types of cache memory is/ are built inside the microprocessor?

I. Level-1 (L1)
II. Level-2 (L2)
III. Level-3 (L3)

A. I only
B. III only
C. I and II
D. II and III

27. Consider the given IPv4 address.

192.100.200.250

The class to which this IP address belongs to is

A. class A.
B. class B.
C. class C.
D. class D.

28. The length of an IPv4 address is 32 bits.

The number of bits required for host ID in **class A** IP address is

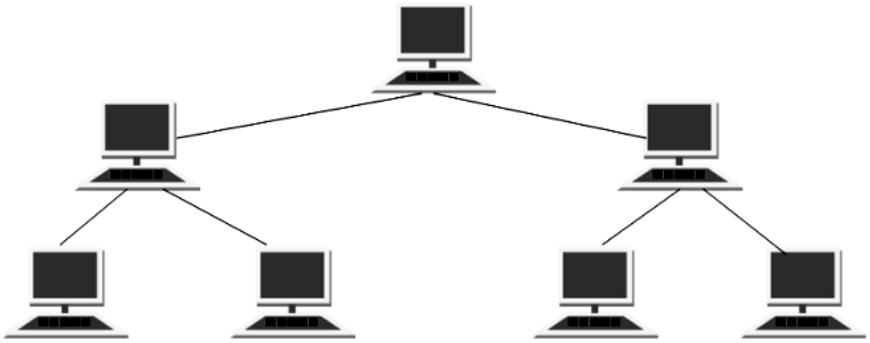
A. 8
B. 16
C. 24
D. 32

29. Three OSI model layers were merged as one layer in TCP/ IP model, i.e. Application layer.

One of these layers was application layer, the other TWO OSI model layers were

A. physical and network.
B. network and transport.
C. session and presentation.
D. transport and presentation.

30. Consider the given picture of a network topology.



The topology that is shown in this picture is

A. bus.
B. star.
C. tree.
D. mesh.

31. When a home antenna receives signals sent by a TV station transmitter, then the mode of transmission is

A. serial.
B. duplex.
C. parallel.
D. simplex.

32. As compared to asynchronous transmission, the synchronous transmission

A. uses start and stop bits.
B. sends data in low speed.
C. inserts gaps between data blocks.
D. sends data in the form of one long stream.

33. The type of network that has the LEAST coverage area is

A. IAN
B. PAN
C. LAN
D. WAN

34. Network created between wireless keyboard and computer system is an example of

A. nano network.
B. body area network.
C. near-me-area network.
D. personal area network.

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35. A logical communication network built on top of existing physical network infrastructures that focuses on communication among wireless devices in proximity is

A. local area network.
B. wide area network.
C. near-me-area network.
D. personal area network.

36. The CORRECT categorisation of protocols according to the given layers of OSI model is

	Application Layer	**Transport Layer**	**Presentation Layer**
A	User Datagram Protocol	File Transfer Protocol	Secure Socket Layer
B	File Transfer Protocol	User Datagram Protocol	Secure Socket Layer
C	Secure Socket Layer	File Transfer Protocol	User Datagram Protocol
D	Secure Socket Layer	User Datagram Protocol	File Transfer Protocol

37. In computer networks, tree topology is designed by combining two topologies that are

A. bus and star.
B. bus and ring.
C. mesh and star.
D. mesh and ring.

38. The set of features that CORRECTLY describes the TCP/ IP model is

A	• It consists of seven layers. • It was developed by ISO.
B	• It consists of four layers. • It is an implementation of OSI model.
C	• It consists of four layers. • It is a reference model.
D	• It consists of seven layers. • It is a layer based model.

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39. Consider the given tasks.

I. Retrieving data from a database
II. Inserting data into a database
III. Deleting Objects from the database
IV. Deleting Records from a database

The tasks performed by data manipulating language of SQL are

A. I, II and III.
B. II, III and IV.
C. I, II and IV.
D. I, III and IV.

40. The collection of related fields in a database table is treated as a single unit. The two names of this unit are

I. Tuple
II. Attribute
III. Column
IV. Record

A. I and II.
B. I and IV.
C. II and III.
D. III and IV.

Use the given information to answer Q.41 and Q.42.

An employee database table has following fields. The Employee_ID is the primary key.

I. Employee_ID
II. Employee_Name
III. Employee_National_ID_Card_Number
IV. Employee_Age
V. Employee_Gender
VI. Employee_Address

41. In this table, the MOST appropriate candidate key is

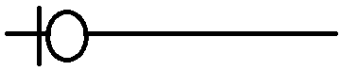
A. II only.
B. III only.
C. IV only.
D. V only.

42. In the given options, the MOST appropriate table field having **number** data type is

A. II only.
B. IV only.
C. V only.
D. VI only.

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43. Consider the given image representing cardinality and modality.



The CORRECT interpretation of cardinality and modality from the given image is

A. cardinality is one and modality is zero.
B. cardinality is zero and modality is one.
C. cardinality is many and modality is zero.
D. cardinality is zero and modality is many.

44. The MS Access query that will fetch records that consist of **CarModel** and **CarPrice** fields from a database table named as **Cars** is

A. Select CarModel, CarPrice From Cars;
B. Select *CarModel, *CarPrice From Cars;
C. Select CarModel AND CarPrice From Cars;
D. Select *(CarModel AND CarPrice) From Cars;

45. The MS Access query that will fetch all the records of clients living in Karachi from a database table named as **Clients** is

A. Select * From Clients Where City = 'Karachi';
B. Select * From Clients Where City = $Karachi;
C. Select * From Karachi Where Table = Clients;
D. Select * From City = Karachi Where Table = Clients;

46. The MS Access query that will fetch all the records of clients having monthly income between 50,000 and 75,000 rupees from a database table named as **Clients** is

A. Select * From Clients Where Income > 50,000 OR Income < 75,000;
B. Select * From Clients Where Income > 50000 AND Income < 75000;
C. Select * From Income = 50000 AND Income = 75000 Where Table = Clients;
D. Select * From Income >= 50000 AND Income =< 75000 Where Table = Clients;

47. Fatima has created a database table named as **Stores**. She wants to insert a column named as **ContactNumber**.

The query that she should use to do this task is

A. Alter Table Stores Add ContactNumber Int;
B. Alter Table Stores Include ContactNumber Int;
C. Alter Stores Table Append ContactNumber Int;
D. Alter Insert ContactNumber Int Where Table = Stores;

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48. Fakhir wants to apply the following restriction on a field of MS Access database table having text data type.

Data entry must be of 4 characters in length and should begin with the letter J

To apply the given restriction, the CORRECT data validation rule will be

A. J$$$
B. #J---#
C. "J???"
D. "J" AND = 4

49. All of the following queries can change records in a database table EXCEPT

A. select query.
B. delete query.
C. update query.
D. append query.

50. Amna has two tables in the same database. She wants to add records from one table to the second table. If the fields of two tables have compatible data types, then the type of query that should be used for adding records is

A. drop query.
B. select query.
C. append query.
D. make table query.

ALTERNATE TO PRACTICAL (ATP: Questions 51-65)

51. A database administrator wants to create one-to-one relationship between two tables named as **X** and **Y**.

He/ she should ensure that a certain criterion is followed. The criterion is that

A. related fields in both tables should be foreign keys.
B. related fields in both tables should be primary keys.
C. the primary field of table **X** is related with secondary field of table **Y**.
D. the secondary field of table **X** is related with primary field of table **Y**.

Use the given table named as Teacher to answer Q.52, Q.53, Q.54, Q.55 and Q.56.

TeacherID	FullName	JoiningDate	Salary
TID230	Umer Abdullah	29/03/2016	55000
TID231	Sana Fatima	17/12/2019	60000
TID232	Mushtaq Ahmed	7/05/2017	52000

Consider the given specifications of this table.

- The **TeacherID** field is the primary key and it should store up to 6 text characters. This field cannot be left empty.
- The **FullName** field should store up to 75 text characters. This field cannot be left empty.
- The **JoiningDate** field should have date data type.
- The **Salary** field should have numeric data type.

52. The SQL query to create this table is

A	CREATE TABLE Teacher (TeacherID text(6) PRIMARY KEY, FullName text(75), JoiningDate date, Salary int);
B	CREATE TABLE Teacher (TeacherID text(6) NOT NULL PRIMARY KEY, FullName text(75) NOT NULL, JoiningDate date, Salary int);
C	CREATE Teacher TABLE (TeacherID text() NOT NULL PRIMARY KEY, FullName text() NOT NULL, JoiningDate date, Salary int);
D	CREATE Teacher TABLE (TeacherID text(6) NOT NULL PRIMARY KEY, JoiningDate date, FullName text(75) NOT NULL, Salary int);

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53. The record having **TeacherID = TID230** needs to be edited as follows:

- Change the JoiningDate to 29/04/2016
- Change the Salary to 65000

The SQL query to do these changes is

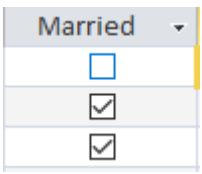
A	UPDATE Teacher SET (JoiningDate = '29/04/2016' AND Salary = 65000) WHERE TeacherID = 'TID230';
B	UPDATE Teacher SET JoiningDate = '29/04/2016' AND SET Salary = 65000 WHERE TeacherID = 'TID230';
C	UPDATE Teacher SET JoiningDate = '29/04/2016', Salary = 65000 WHERE TeacherID = 'TID230';
D	UPDATE Teacher SET JoiningDate = '29/04/2016', Salary = 65000 WHERE TeacherId = <TID230>;

54. If descending sort option is applied on one of the fields of this table, then the record of Sana Fatima becomes first, the record of Mushtaq Ahmed becomes second and the record of Umer Abdullah becomes third in the table.

The field on which the descending sort option is applied will be

A. TeacherID
B. FullName
C. JoiningDate
D. Salary

55. The following picture shows a newly added field in the datasheet view of the given table.



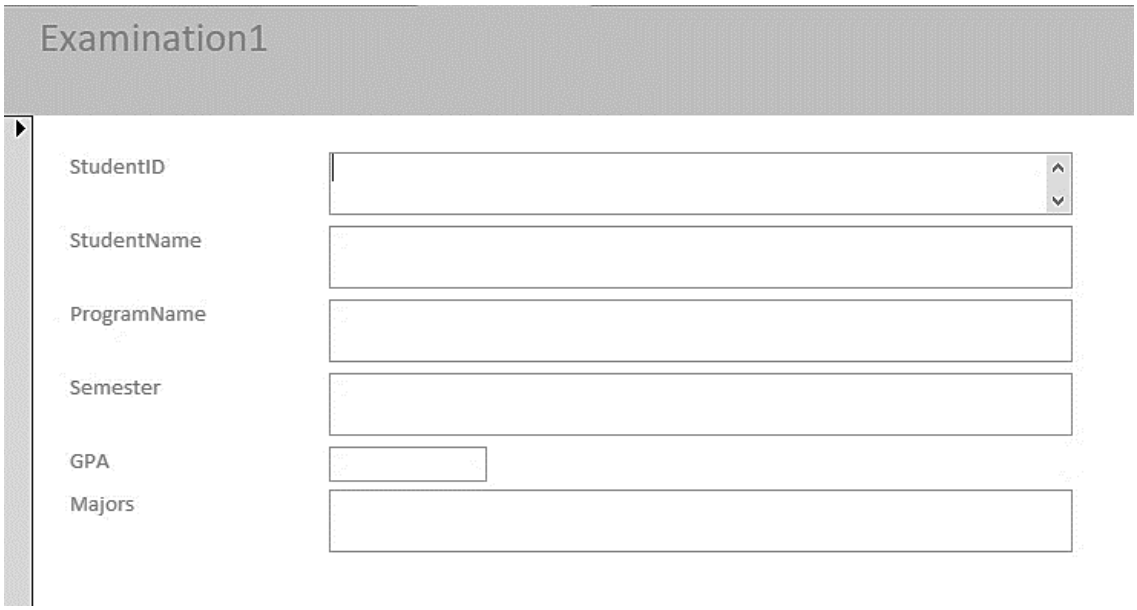
The data type of this field is

A. Memo
B. Yes/ No
C. Hyperlink
D. OLE Object

56. The SQL query in MS Access to delete this table is

A. Delete Teacher Table;
B. Delete Table Teacher;
C. Drop Teacher Table;
D. Drop Table Teacher;

57. Consider the given form.



The layout of the form is

A. tabular.
B. justified.
C. datasheet.
D. columnar.

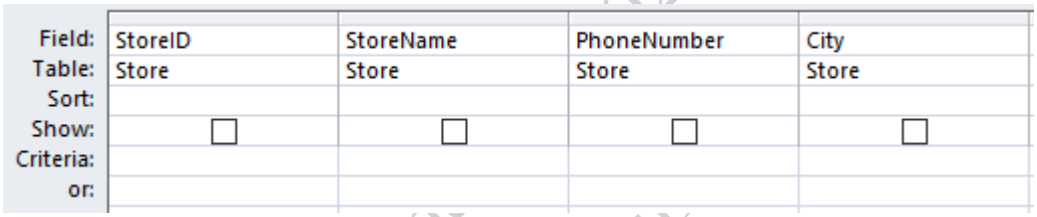
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Use the given table named as Store to answer Q.58, Q.59, Q.60, Q.61 and Q.62.

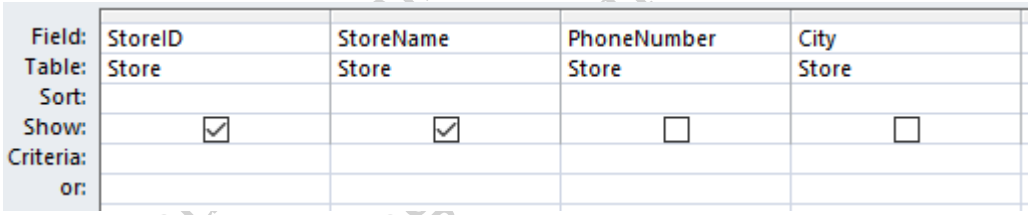
StoreID	StoreName	PhoneNumber	City
S150	Super Store	11223344	Hyderabad
S151	Best Store	22334455	Karachi
S152	Galaxy Store	33445566	Lahore
S153	Diamond Store	44556677	Karachi
S154	Lucky Store	55667788	Gilgit

58. The query created in design view to display all fields and records of the table is

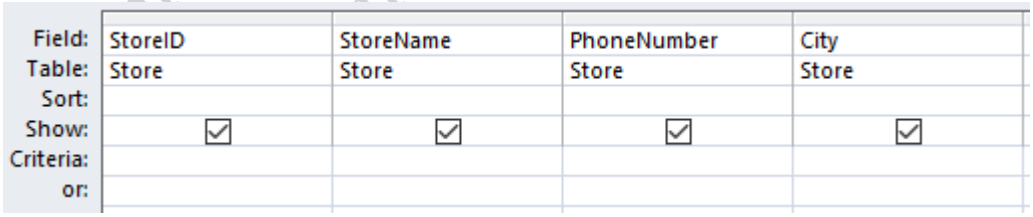
A



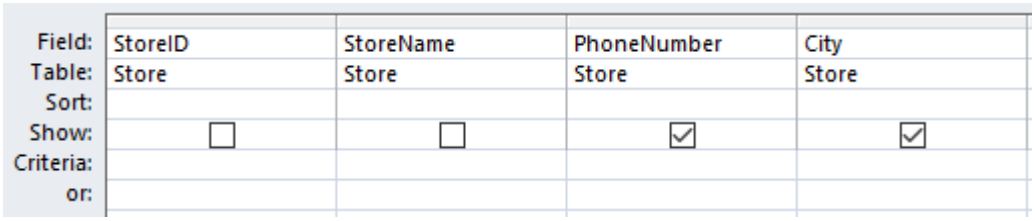
B



C

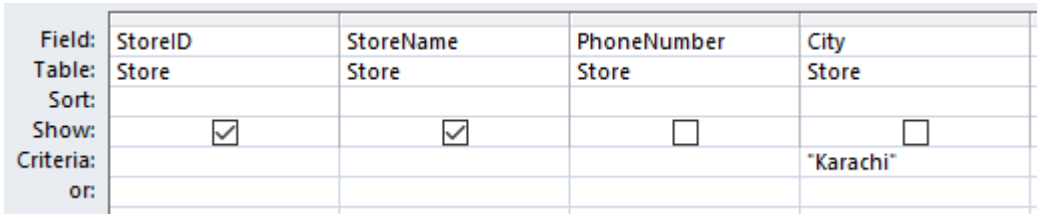


D

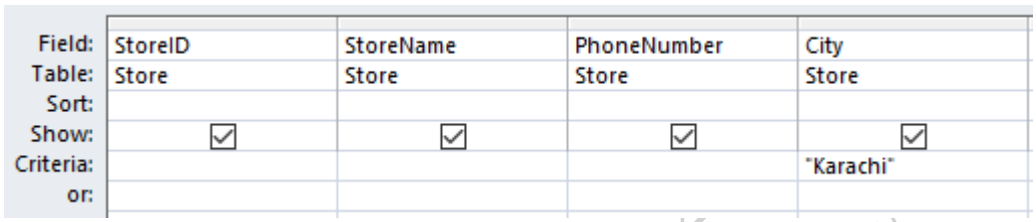


59. The query created in design view to only display the store ID and store name of all stores in Karachi is

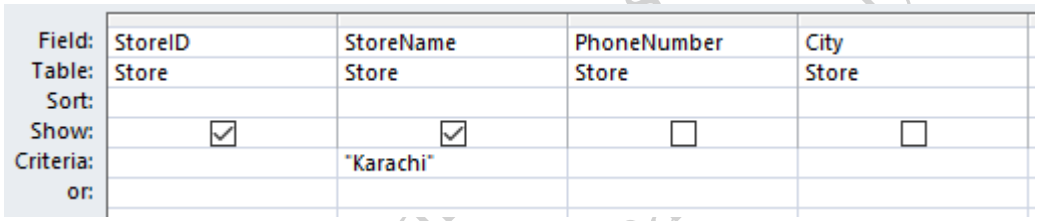
A



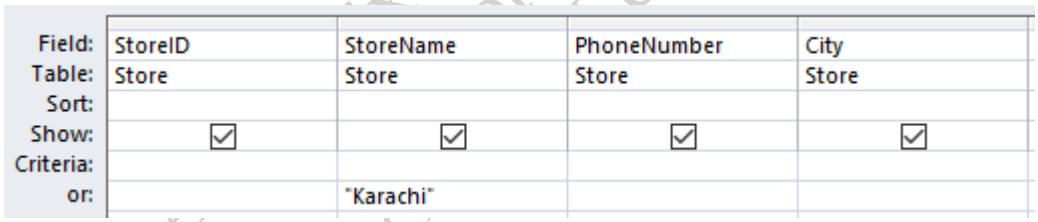
B



C

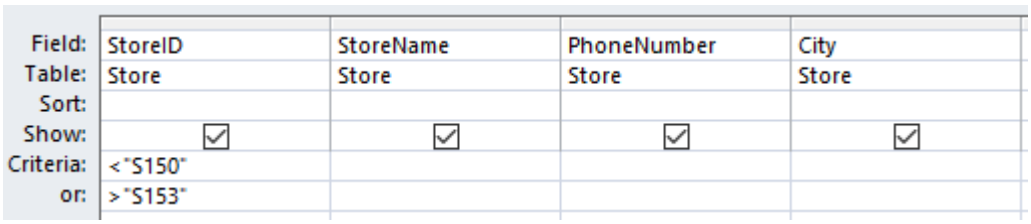


D

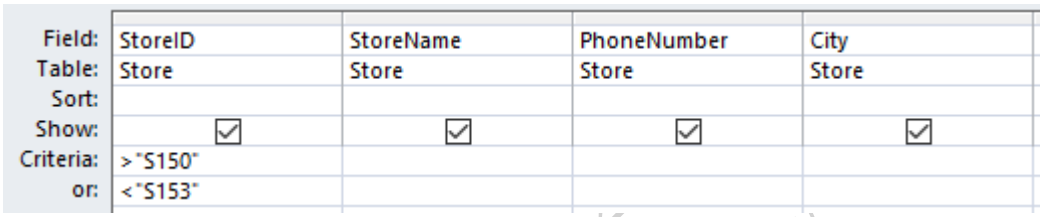


60. The query created in design view to display the records of all those stores whose store ID is between S150 and S153 is

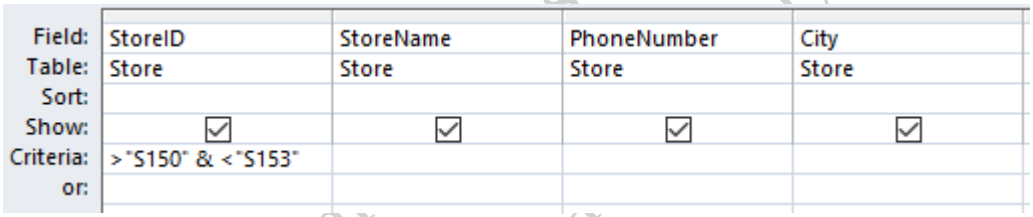
A



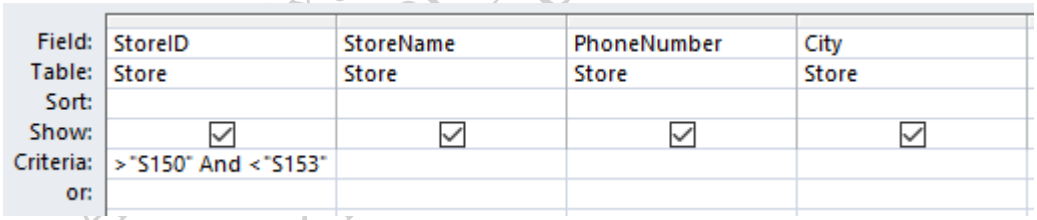
B



C



D



61. The query created in design view to change the name of Best Store to Silver Store is

A

Field:	StoreID	StoreName	PhoneNumber	City
Table:	Store	Store	Store	Store
Update To:		"Silver Store"		
Criteria:		"S151"		
or:				

B

Field:	StoreID	StoreName	PhoneNumber	City
Table:	Store	Store	Store	Store
Update To:		"Silver Store"		
Criteria:	"S151"			
or:				

C

Field:	StoreID	StoreName	PhoneNumber	City
Table:	Store	Store	Store	Store
Update To:	"Silver Store"			
Criteria:				
or:	"S151"			

D

Field:	StoreID	StoreName	PhoneNumber	City
Table:	Store	Store	Store	Store
Sort:				
Show:	☑	☑	☑	☑
Criteria:		"Silver Store"		
or:	"S151"			

62. The query created in design view to remove the records of stores having store IDs S151 and S152 is

A

Field:	StoreID	StoreName	PhoneNumber	City
Table:	Store	Store	Store	Store
Delete:	Where	Where	Where	Where
Criteria:	"S151"			
or:		"S152"		

B

Field:	StoreID	StoreName	PhoneNumber	City
Table:	Store	Store	Store	Store
Delete:	Where	Where	Where	Where
Criteria:	"S151" And "S152"			
or:				

C

Field:	StoreID	StoreName	PhoneNumber	City
Table:	Store	Store	Store	Store
Delete:	Where	Where	Where	Where
Criteria:			22334455 And 33445566	

D

Field:	StoreID	StoreName	PhoneNumber	City
Table:	Store	Store	Store	Store
Delete:	Where	Where	Where	Where
Criteria:			22334455	
or:			33445566	

63. To produce an MS Access report, the data is fetched with the help of

A. tables and queries.
B. forms and queries.
C. tables and relationships.
D. forms and relationships.

64. Usman wants to change the data type of a field in a table. The type of table view he should use for doing this task is

A. Design View.
B. Datasheet View.
C. PivotTable View.
D. PivotChart View.

65. The SQL query to select all the records and attributes from a table named as **Cars** is

A. Select All From Cars Table;
B. Select * From Table Cars;
C. Select All From Cars;
D. Select * From Cars;

END OF PAPER

Please use this page for rough work

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